Stochastic Perturbed Proximal Augmented Lagrangian with Fixed Penalty for Nonconvex Constrained Optimization

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1 Introduction

We consider the following nonconvex stochastic optimization problem with equality and inequality constraints:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathcal{X}} \quad & F(\boldsymbol{x}) := f(\boldsymbol{x}) + g(\boldsymbol{x}), \\ \text{s.t.} \quad & \boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x}) = \mathbf{0}, \\ & \boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x}) \leq \mathbf{0}, \end{aligned} \tag{\textbf{P}}$$

where $f(\boldsymbol{x}) = \mathbb{E}_{\xi}[\mathbf{f}(\boldsymbol{x}, \xi)]$ is a smooth nonconvex function with ξ a random variable on a probability space Ξ independent of $\boldsymbol{x}$, and $g : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is a proper closed proximal-friendly function. The mappings $\boldsymbol{c}_{\mathcal{E}} : \mathbb{R}^n \to \mathbb{R}^{|\mathcal{E}|}$ and $\boldsymbol{c}_{\mathcal{I}} : \mathbb{R}^n \to \mathbb{R}^{|\mathcal{I}|}$ are continuously differentiable. The set $\mathcal{X} \subseteq \mathbb{R}^n$ is assumed to be nonempty and closed. The feasible set is assumed to be nonempty and F is lower bounded by $F^* > -\infty$. Exact evaluation of ∇f is often expensive or impractical, and thus we assume only access to stochastic gradients of f at queried points.

1.1 Related Work

Deterministic nonconvex constrained optimization. For deterministic nonconvex problems with equality and inequality constraints, first-order methods with non-asymptotic complexity guarantees have been studied extensively. Proximal point methods [4, 20] achieve $\mathcal{O}(\varepsilon^{-3})$ iteration complexity for smooth objectives via strongly convex subproblems, and Boob et al. [5] further attain $\mathcal{O}(\varepsilon^{-2})$ by constructing quadratic models with adaptive constraint levels. Among penalty and augmented Lagrangian methods (ALMs), most approaches require the penalty parameter to grow as $\Omega(\varepsilon^{-1})$, yielding $\mathcal{O}(\varepsilon^{-3})$ iteration complexity [15, 13]. Xie and Wright [24] achieve $\mathcal{O}(\varepsilon^{-2})$ outer-loop complexity with a proximal ALM under a constant merit parameter, but the algorithm is double-loop and the merit parameter must satisfy $\rho = \Omega(\varepsilon^{-\eta})$ for some $\eta \in (0, 1)$, making it ε-dependent in theory. SQP methods [9, 6] also attain $\mathcal{O}(\varepsilon^{-2})$ complexity.

Stochastic nonconvex constrained optimization. We organize the literature by the two canonical settings: *semi-stochastic* (stochastic objective, deterministic constraints) and *fully stochastic* (both objective and constraints involve stochastic oracles).

Semi-stochastic problems. Without additional structure on the noise, proximal point [4, 20], penalty [14], and SQP [9] methods all achieve $\mathcal{O}(\varepsilon^{-4})$ oracle complexity in stochastic gradient evaluations, matching the $\mathcal{O}(\varepsilon^{-4})$ lower bound for unconstrained stochastic optimization [3]. Under a mean-squared smoothness (MSS) condition, variance reduction techniques—STORM [10] and SPIDER [11]—improve the complexity to $\mathcal{O}(\varepsilon^{-3})$. Representative results include the variance-reduced linearized ALM of Shi et al. [23], the truncated variance reduction approach of Lu et al. [18], and the inexact ALM of Li et al. [15], each attaining $\mathcal{O}(\varepsilon^{-3})$ oracle complexity under MSS for nonlinear constraints.

Fully stochastic problems. When constraint evaluations are also stochastic, complexity results are uniformly higher. Proximal point methods [4, 20] incur $\mathcal{O}(\varepsilon^{-6})$ total oracle complexity due to the cost of solving stochastic strongly convex subproblems. Penalty-based ALMs employing STORM-type variance reduction [1, 7] face a large merit parameter $\beta = \Theta(\varepsilon^{-1})$, which amplifies the variance of the stochastic gradient and yields $\mathcal{O}(\varepsilon^{-5})$ complexity even under MSS. Exact penalty methods combined with SPIDER [17] achieve $\mathcal{O}(\varepsilon^{-4})$ (sub)gradient and $\mathcal{O}(\varepsilon^{-6})$ constraint-evaluation complexity for nonsmooth problems. Cui et al. [8] recently propose an adaptive exact penalty method for smooth *equality*-constrained problems, obtaining $\mathcal{O}(\varepsilon^{-3})$ gradient and $\mathcal{O}(\varepsilon^{-5})$ constraint-evaluation complexity; however, the analysis is confined to equality constraints and relies on an exact penalty reformulation. Stochastic SQP for equality-constrained fully stochastic problems [21] currently achieves only $\mathcal{O}(\varepsilon^{-8})$.

ALMs with fixed penalty parameter. A fundamental challenge in ALM design for nonlinear constraints is that standard convergence analysis requires the penalty parameter to scale as $\Omega(\varepsilon^{-1})$. This growth inflates the Lipschitz constant of the augmented Lagrangian and forces the step size to shrink proportionally, degrading the stochastic oracle complexity. For polyhedral (linear) constraints, the Luo–Pang global error bound [19] decouples feasibility control from the penalty magnitude. Exploiting this structure, Zhang and Luo [25] achieve $\mathcal{O}(\varepsilon^{-2})$ iteration complexity for deterministic nonconvex problems under a genuinely constant penalty. In the stochastic setting, Alacaoglu and Wright [2] combine a fixed penalty with variance reduction to attain the optimal $\mathcal{O}(\varepsilon^{-3})$ sample complexity for linear equality constraints; for nonlinear constraints, their single-loop algorithm requires an increasing penalty and achieves only $\mathcal{O}(\varepsilon^{-4})$. Huang et al. [12] extend the fixed-penalty result to linear inequality constraints via Moreau envelope smoothing, also reaching $\mathcal{O}(\varepsilon^{-3})$. For general nonlinear constraints, existing stochastic methods either require an increasing penalty parameter [16, 2], yielding $\mathcal{O}(\varepsilon^{-4})$ oracle complexity, or introduce dual momentum to partially compensate, achieving $\mathcal{O}(\varepsilon^{-3.5})$ [22]. To date, no single-loop ALM with a genuinely constant penalty parameter has been shown to achieve $\mathcal{O}(\varepsilon^{-3})$ oracle complexity for nonlinear nonconvex constraints in the stochastic setting.

The proposed Perturbed ALM addresses this gap. By introducing a dual regularization term $-\frac{\beta}{2}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2$ into the augmented Lagrangian, the effective penalty parameter reduces to $\rho = \alpha/(1+\alpha\beta)$, which is *independent* of ε for any fixed $\alpha, \beta > 0$. This perturbation simultaneously enables a *single-loop* primal–dual update and achieves $\mathcal{O}(\varepsilon^{-3})$ stochastic oracle complexity for nonlinear nonconvex constrained optimization, improving over the existing $\mathcal{O}(\varepsilon^{-4})$ barrier for this setting.

1.2 Our Contributions

Can we design algorithms to solve problems of the form equation 1 with an improved iteration complexity bound and efficient update rules?

1.3 Notation

Throughout this paper, $\|\cdot\|$ denotes the Euclidean norm and $\langle\cdot,\cdot\rangle$ denotes the Euclidean inner product. We use $\mathbb{E}_\xi[\cdot]$ for expectation with respect to ξ. Let $[p] := \{1,\dots,p\}$, $\mathbb{N} := \{0,1,2,\dots\}$, and $\overline{\mathbb{R}} := \mathbb{R}\cup\{+\infty\}$. For $A \in \mathbb{R}^{m\times n}$, $\|A\|$ and $\sigma_{\min}(A)$ denote the largest and smallest singular values of A, respectively. For $X \subseteq \mathbb{R}^n$, δ_X denotes its indicator function and $\operatorname{dist}(x,X) := \inf_{y\in X}\|x-y\|$. For complexity notation, we write $K = \mathcal{O}(\varepsilon)$ if $K \le B\varepsilon$ for some $B > 0$, and $K = \Theta(\varepsilon)$ if $b\varepsilon \le K \le B\varepsilon$ for some constants $0 < b < B < +\infty$.

1.4 Organization

The rest of the paper is organized as follows. Section 2 introduces the assumptions and the stochastic KKT criterion. Section 3 presents the perturbed proximal ALM formulation and the single-loop algorithm. Section 4 establishes stochastic complexity and residual bounds. Section 5 reports numerical experiments. Additional technical lemmas are provided in Appendix A.

2 Preliminaries and Assumptions

2.1 Main Assumptions and stationarity

In this subsection, we formally state our assumptions for Problem (**P**) and the stationarity for problem

Assumption 1 *Function $f(\cdot)$ is L_f-smooth over $\mathcal{X}$. The mappings $\boldsymbol{c}_{\mathcal{E}} : \mathbb{R}^n \to \mathbb{R}^{|\mathcal{E}|}$ and $\boldsymbol{c}_{\mathcal{I}} : \mathbb{R}^n \to \mathbb{R}^{|\mathcal{I}|}$ are continuously differentiable on $\mathcal{X}$. There exist constants $C_{\mathcal{E}}, C_{\mathcal{I}}, G_{\mathcal{E}}, G_{\mathcal{I}}, L_{\mathcal{E}}, L_{\mathcal{I}} > 0$ such that for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}$,*

$$\begin{aligned} \|\boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x})\| &\le C_{\mathcal{E}}, & \|\boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x})\| &\le C_{\mathcal{I}}, \\ \|J_{\boldsymbol{c}_{\mathcal{E}}}(\boldsymbol{x})\| &\le G_{\mathcal{E}}, & \|J_{\boldsymbol{c}_{\mathcal{I}}}(\boldsymbol{x})\| &\le G_{\mathcal{I}}, \\ \|J_{\boldsymbol{c}_{\mathcal{E}}}(\boldsymbol{x}) - J_{\boldsymbol{c}_{\mathcal{E}}}(\boldsymbol{y})\| &\le L_{\mathcal{E}}\|\boldsymbol{x}-\boldsymbol{y}\|, & \|J_{\boldsymbol{c}_{\mathcal{I}}}(\boldsymbol{x}) - J_{\boldsymbol{c}_{\mathcal{I}}}(\boldsymbol{y})\| &\le L_{\mathcal{I}}\|\boldsymbol{x}-\boldsymbol{y}\|. \end{aligned}$$

Define the aggregated constraint mapping $\boldsymbol{c}(\boldsymbol{x}) := [\boldsymbol{c}_{\mathcal{E}}(\boldsymbol{x})^\top, \boldsymbol{c}_{\mathcal{I}}(\boldsymbol{x})^\top]^\top$. Then there exist $C, G, L_c > 0$ such that

$$\|\boldsymbol{c}(\boldsymbol{x})\| \le C, \qquad \|J_{\boldsymbol{c}}(\boldsymbol{x})\| \le G, \qquad \|J_{\boldsymbol{c}}(\boldsymbol{x}) - J_{\boldsymbol{c}}(\boldsymbol{y})\| \le L_c\|\boldsymbol{x}-\boldsymbol{y}\|, \quad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}. \tag{2.1}$$

Assumption 2 *There exists a $\sigma > 0$ such that*

$$\begin{aligned} &\mathbb{E}_\xi[\nabla \mathbf{f}(\boldsymbol{x}, \boldsymbol{\xi})] = \nabla f(\boldsymbol{x}), \\ &\mathbb{E}_\xi[\|\nabla \mathbf{f}(\boldsymbol{x}, \boldsymbol{\xi}) - \nabla f(\boldsymbol{x})\|^2] \le \sigma^2, \quad \forall \boldsymbol{x} \in \mathcal{X}. \end{aligned}$$

Assumption 3 (Boundedness of Multipliers) *The sequence of dual variables $\{\boldsymbol{\lambda}^k\}$ generated by Algorithm 1 is contained in a convex compact set $\Lambda \subset \mathbb{R}^m$, i.e., there exists a constant $B_\lambda > 0$ such that*

$$\|\boldsymbol{\lambda}^k\| \le B_\lambda, \quad \forall\, k \ge 0. \tag{2.2}$$

Before proceeding, we introduce an additional standard assumption, also known as the mean-squared smoothness assumption, widely used in variance reduction methods.

Assumption 4 *For almost every $\boldsymbol{\xi} \in \Xi$, $f(\cdot, \boldsymbol{\xi})$ is differentiable and for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}$,*

$$\mathbb{E}_{\boldsymbol{\xi}}\left[\|\nabla f(\boldsymbol{x}; \boldsymbol{\xi}) - \nabla f(\boldsymbol{y}; \boldsymbol{\xi})\|^2\right] \le L_f^2\|\boldsymbol{x}-\boldsymbol{y}\|^2.$$

2.2 Stochastic KKT Point

Definition 1 (Stochastic KKT point) *A point $\boldsymbol{x}^* \in \mathcal{X}$ is called a* stationary point *of Problem (**P**) if there exists $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ such that the KKT conditions are satisfied:*

$$\mathbf{0} \in \nabla f(\boldsymbol{x}^*) + \partial g(\boldsymbol{x}^*) + J_{\boldsymbol{c}}(\boldsymbol{x}^*)^\top \boldsymbol{\lambda}^* + \mathcal{N}_{\mathcal{X}}(\boldsymbol{x}^*), \qquad \boldsymbol{c}(\boldsymbol{x}^*) = \mathbf{0}. \tag{2.3}$$

3 Methodology and Algorithm

We introduce a residual variable $\boldsymbol{z}$ and rewrite (**P**) as

$$\min_{\boldsymbol{x} \in \mathcal{X}, \boldsymbol{z}} \; f(\boldsymbol{x}) + g(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{c}(\boldsymbol{x}) = \boldsymbol{z}, \quad \boldsymbol{z} = \mathbf{0}. \tag{3.1}$$

Following [], we define the perturbed proximal Lagrangian

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{\lambda}, \boldsymbol{\mu}) := f(\boldsymbol{x}) + g(\boldsymbol{x}) + \langle \boldsymbol{\lambda}, \boldsymbol{c}(\boldsymbol{x}) - \boldsymbol{z}\rangle + \langle \boldsymbol{\mu}, \boldsymbol{z}\rangle + \frac{\alpha}{2}\|\boldsymbol{z}\|^2 - \frac{\beta}{2}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2 + \frac{\rho}{2}\|\boldsymbol{c}(\boldsymbol{x})\|^2, \tag{3.2}$$

where $\boldsymbol{\lambda}, \boldsymbol{\mu} \in \mathbb{R}^m$ are dual variables. Eliminating $\boldsymbol{z}$ via $\boldsymbol{z}^*(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \frac{\boldsymbol{\lambda} - \boldsymbol{\mu}}{\alpha}$ yields the reduced PPAL:

$$\mathcal{L}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \underbrace{f(\boldsymbol{x}) + \langle \boldsymbol{\lambda}, \boldsymbol{c}(\boldsymbol{x})\rangle + \frac{\rho}{2}\|\boldsymbol{c}(\boldsymbol{x})\|^2}_{\varphi_\rho(\boldsymbol{x}, \boldsymbol{\lambda})} + g(\boldsymbol{x}) - \frac{1}{2\rho}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2, \tag{3.3}$$

where $\rho := \frac{\alpha}{1+\alpha\beta} > 0$.

3.1 Algorithm

Stochastic gradient estimator. We replace $\nabla f(\boldsymbol{x}^k)$ with the STORM variance-reduced estimator:

$$\boldsymbol{d}^k = \nabla \mathbf{f}(\boldsymbol{x}^k, \xi^k) + (1-\gamma_{k-1})\big(\boldsymbol{d}^{k-1} - \nabla \mathbf{f}(\boldsymbol{x}^{k-1}, \xi^k)\big), \tag{3.4}$$

where ξ^k is an independent sample drawn at iteration k.

Update of $\boldsymbol{x}$.

$$\boldsymbol{x}^{k+1} \in \operatorname*{argmin}_{\boldsymbol{x}\in\mathcal{X}} \left\{ g(\boldsymbol{x}) + \left\langle \hat{\nabla}_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k),\ \boldsymbol{x} - \boldsymbol{x}^k \right\rangle + \frac{1}{2\eta}\|\boldsymbol{x} - \boldsymbol{x}^k\|^2 \right\}, \tag{3.5}$$

where $\hat{\nabla}_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k) := \boldsymbol{d}^k + J_{\boldsymbol{c}}(\boldsymbol{x}^k)^\top \left(\boldsymbol{\lambda}^k + \rho\, \boldsymbol{c}(\boldsymbol{x}^k)\right)$.

Update of $\boldsymbol{\mu}$.

$$\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \theta_k\big(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\big), \tag{3.6}$$

with a diminishing step-size

$$\theta_k = \min\left\{\theta_0,\ \frac{\rho\delta_k}{\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 + 1}\right\}, \tag{3.7}$$

and δ_k satisfies

$$\delta_0 \in (0,1], \qquad \lim_{k\to\infty} \delta_k = 0, \qquad \text{and} \qquad \sum_{k=0}^{\infty} \delta_k = +\infty. \tag{3.8}$$

Update of $\boldsymbol{\lambda}$.

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\mu}^{k+1} + \rho\, \boldsymbol{c}(\boldsymbol{x}^{k+1}). \tag{3.9}$$

Update of $\boldsymbol{z}$.

$$\boldsymbol{z}^{k+1} = \frac{1}{\alpha}\big(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\big). \tag{3.10}$$

Algorithm 1 Perturbed Linearized ALM

1: **Input:** $(\boldsymbol{x}^0, \boldsymbol{\lambda}^0, \boldsymbol{\mu}^0)$, parameters $\eta > 0$, $\alpha > 0$, $\beta > 0$ and γ, sequence $\{\delta_k\}$ with $\delta_k \to 0$, $\sum_k \delta_k = +\infty$.
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Update $\boldsymbol{x}^{k+1}$ by (3.5).
4: Update $\boldsymbol{\mu}^{k+1}$ by (3.6).
5: Update $\boldsymbol{\lambda}^{k+1}$ by (3.9).
6: Update $\boldsymbol{z}^{k+1}$ by (3.10).
7: **end for**
8: **Output:** $\boldsymbol{x}^R$ (randomly selected iterate).

4 Stochastic Complexity Analysis

Lemma 1 (Smoothness of φ_ρ) *Under Assumptions 1 and 3, the function $\varphi_\rho(\cdot, \boldsymbol{\lambda})$ is L_φ-smooth for any $\boldsymbol{\lambda}$ with $\|\boldsymbol{\lambda}\| \le B_\lambda$, i.e.,*

$$\|\nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) - \nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{y}, \boldsymbol{\lambda})\| \le L_\varphi \|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}, \tag{4.1}$$

where

$$L_\varphi := L_f + L_c(B_\lambda + \rho C) + \rho G^2. \tag{4.2}$$

Lemma 2 (Basic relations on $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$) *Let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be the sequence generated by Algorithm 1. Under Assumption 1, the following relations hold for all $k \geq 0$:*

$$\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\|^2 = \theta_k^2 \|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 \leq \frac{\rho^2 \delta_k^2}{4}, \tag{4.3}$$

$$\theta_k \|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2 \leq \rho \delta_k, \tag{4.4}$$

$$\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\lambda}^k\|^2 = (1-\theta_k)^2 \|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k\|^2, \tag{4.5}$$

$$\mathbb{E}[\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\|^2] \leq 2\rho^2 L_c^2 \, \mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2] + \frac{\rho^2 \delta_k^2}{2}. \tag{4.6}$$

Lemma 3 (Sufficient Descent) *Let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be the sequence generated by Algorithm 1. Under Assumptions 1, 2, and 3, the following inequality holds for all $k \geq 0$:*

$$\mathbb{E}\big[\mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}, \boldsymbol{\mu}^{k+1}) \,\big|\, \mathcal{F}_k\big] \;\leq\; \mathcal{L}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) - C_1 \, \mathbb{E}\big[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \,\big|\, \mathcal{F}_k\big] + \frac{\eta}{2}\|\boldsymbol{e}^k\|^2 + \hat{\delta}_k, \tag{4.7}$$

where

$$C_1 := \frac{1}{2\eta} - \frac{L_\varphi}{2} - \rho L_c^2 > 0, \qquad \hat{\delta}_k := \frac{\rho \delta_k^2}{4} + \delta_k. \tag{4.8}$$

Furthermore, rearranging (4.7) gives

$$C_1 \, \mathbb{E}\big[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k\big] \leq \mathcal{L}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) - \mathbb{E}\big[\mathcal{L}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}, \boldsymbol{\mu}^{k+1}) \mid \mathcal{F}_k\big] + \frac{\eta}{2}\|\boldsymbol{e}^k\|^2 + \hat{\delta}_k. \tag{4.9}$$

Lemma 4 (Primal Stationarity Residual Bound) *Let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be the sequence generated by Algorithm 1. Under Assumptions 1 and 2, define the stationarity residual*

$$\boldsymbol{\zeta}_{\boldsymbol{x}}^{k+1} \in \nabla_{\boldsymbol{x}} \varphi_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}) + \partial g(\boldsymbol{x}^{k+1}) = \partial_{\boldsymbol{x}} \mathcal{L}_\rho(\boldsymbol{w}^k). \tag{4.10}$$

Then for all $k \geq 0$, Lemma 2 (4.6),

$$\mathbb{E}\big[\|\boldsymbol{\zeta}_{\boldsymbol{x}}^{k+1}\|^2\big] \;\leq\; 4\left[\left(L_\varphi + \frac{1}{\eta}\right)^2 + 2G^2\rho^2 L_c^2\right] \mathbb{E}\big[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2\big] \;+\; 4\,\mathbb{E}\big[\|\boldsymbol{e}^k\|^2\big] \;+\; 2G^2\rho^2\delta_k^2. \tag{4.11}$$

where $\boldsymbol{e}^k := \boldsymbol{d}^k - \nabla f(\boldsymbol{x}^k)$ is the STORM estimation error.

Lemma 5 (Recursive Gradient Error) *Suppose that Assumptions 1, 2, and 4 hold. Then, for any $k \geq 0$, it holds that*

$$\mathbb{E}\big[\|\boldsymbol{e}^{k+1}\|^2\big] \leq (1-\gamma_k)^2 \, \mathbb{E}\big[\|\boldsymbol{e}^k\|^2\big] + 2(1-\gamma_k)^2 L_f^2 \, \mathbb{E}\big[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2\big] + 2\gamma_k^2 \sigma^2. \tag{4.12}$$

Lemma 6 *Under Assumptions 1–4, let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\}$ be generated by Algorithm 1 with non-increasing step sizes $\{\eta_k\}$ satisfying*

$$(1 - L_{c,\varphi}\eta_k)^{-1} \leq 2 \tag{4.13}$$

where $L_{c,\varphi} = L_\varphi + 2\rho L_c^2$. Then for any $K \geq 1$,

$$\frac{1}{K}\sum_{k=0}^{K-1} \gamma_k \, \mathbb{E}\big[\|\boldsymbol{e}^k\|^2\big] \leq \frac{16\eta_1 L_f^2 \big(\mathcal{L}_\rho^0 - \mathcal{L}_\rho^*\big)}{K} + \frac{2\,\mathbb{E}\big[\|\boldsymbol{e}^1\|^2\big]}{K} + \frac{16\eta_1 L_f^2}{K}\sum_{k=0}^{K-1} \hat{\delta}_k + \frac{1}{K}\sum_{k=0}^{K-1} 4\gamma_k^2\sigma^2, \tag{4.14}$$

where $\mathcal{L}_\rho^* \geq F^*$ is a lower bound on $\mathcal{L}_\rho$.

Lemma 7 (KKT residual stationarity)

$$\frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}\big[\|\zeta_x^{k+1}\|^2\big]$$

$$\leq \left(\frac{\eta_1 L_f^2}{\gamma_K \cdot K} + \frac{1}{\eta_k \cdot K}\right)\big(\mathcal{L}_\rho^0 - \mathcal{L}^*\big) + \frac{2\,\mathbb{E}\big[\|\mathbf{e}^1\|^2\big]}{\gamma_K \cdot K} + \frac{\sum_{k=0}^{K-1} 4\gamma_k^2\sigma^2}{\gamma_K \cdot K} + \frac{1}{K}\sum_{k=0}^{K-1}\delta_k + \left[\frac{1}{\eta_K \cdot K} + \frac{16\eta_1 L_f^2}{\gamma_K \cdot K}\right]\sum_{k=0}^{K-1}\hat{\delta}_k \tag{4.15}$$

Lemma 8 (Violation Bound) *Under the same conditions as Lemma 3, with $\delta_k = c/(k+1)$ for some $c > 0$, we have*

$$\frac{1}{T}\sum_{k=0}^{T-1}\mathbb{E}\left[\|\boldsymbol{c}(\boldsymbol{x}^{k+1})\|^2\right] = \mathcal{O}\left(\frac{\log T}{T}\right) \tag{4.16}$$

Proof 1 *Let $r_k := \boldsymbol{\lambda}^k - \boldsymbol{\mu}^k$. By the $\boldsymbol{\lambda}$-update (3.9), we have the identity*

$$\|r_{k+1}\|^2 = \|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2 = \rho^2\|\boldsymbol{c}(\boldsymbol{x}^{k+1})\|^2, \tag{4.17}$$

so it suffices to bound $\frac{1}{T}\sum_{k=0}^{T-1}\mathbb{E}[\|r_{k+1}\|^2]$.

Step 1: Derive the recursion for $\|r_{k+1}\|^2$. *By the triangle inequality and $(a+b)^2 \le 2(a^2+b^2)$,*

$$\|r_{k+1}\|^2 \le 2\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\|^2 + 2\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^{k+1}\|^2. \tag{4.18}$$

Applying Lemma 2 (4.5) to the second term and (4.6) to the first:

$$\|\boldsymbol{\lambda}^k - \boldsymbol{\mu}^{k+1}\|^2 = (1-\theta_k)^2\|r_k\|^2,$$
$$\mathbb{E}[\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\|^2 \mid \mathcal{F}_k] \le 2\rho^2 L_c^2\,\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k] + \frac{\rho^2\delta_k^2}{2}.$$

Substituting into (4.18) and using $(1-\theta_k)^2 \le (1-\theta_k)$ (since $\theta_k \in (0,1]$):

$$\mathbb{E}[\|r_{k+1}\|^2 \mid \mathcal{F}_k] \le 2(1-\theta_k)\|r_k\|^2 + 4\rho^2 L_c^2\,\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k] + \rho^2\delta_k^2. \tag{4.19}$$

Step 2: Sum the recursion. *Rearranging (4.19):*

$$2\theta_k\|r_k\|^2 \le 2\|r_k\|^2 - \mathbb{E}[\|r_{k+1}\|^2 \mid \mathcal{F}_k] + 4\rho^2 L_c^2\,\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2 \mid \mathcal{F}_k] + \rho^2\delta_k^2. \tag{4.20}$$

Taking total expectation and summing over $k = 0, \dots, T-1$:

$$2\sum_{k=0}^{T-1}\theta_k\,\mathbb{E}[\|r_k\|^2] \le 2\sum_{k=0}^{T-1}\mathbb{E}[\|r_k\|^2] - \sum_{k=0}^{T-1}\mathbb{E}[\|r_{k+1}\|^2] + 4\rho^2 L_c^2\sum_{k=0}^{T-1}\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2] + \rho^2\sum_{k=0}^{T-1}\delta_k^2. \tag{4.21}$$

Note that the first two sums on the right telescope:

$$2\sum_{k=0}^{T-1}\mathbb{E}[\|r_k\|^2] - \sum_{k=0}^{T-1}\mathbb{E}[\|r_{k+1}\|^2] = \sum_{k=0}^{T-1}\mathbb{E}[\|r_k\|^2] - \sum_{k=1}^{T-1}\mathbb{E}[\|r_k\|^2] + \mathbb{E}[\|r_0\|^2] - \mathbb{E}[\|r_T\|^2] \le \mathbb{E}[\|r_0\|^2]. \tag{4.22}$$

Substituting (4.22) into (4.21):

$$2\sum_{k=0}^{T-1}\theta_k\,\mathbb{E}[\|r_k\|^2] \le \|r_0\|^2 + 4\rho^2 L_c^2\sum_{k=0}^{T-1}\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2] + \rho^2\sum_{k=0}^{T-1}\delta_k^2. \tag{4.23}$$

Dividing both sides of (4.23) by T:

$$\frac{2}{T}\sum_{k=0}^{T-1}\theta_k\,\mathbb{E}[\|r_k\|^2] \le \frac{\|r_0\|^2}{T} + \frac{4\rho^2 L_c^2}{T}\sum_{k=0}^{T-1}\mathbb{E}[\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\|^2] + \frac{\rho^2}{T}\sum_{k=0}^{T-1}\delta_k^2. \tag{4.24}$$

5 Numerical Experiments

A Supported Lemmas

Lemma 9 (Vector Azuma-Hoeffding Inequality) *Let $\{\boldsymbol{v}^k\}_{k=1}^K$ be a vector-valued martingale difference sequence adapted to filtration $\{\mathcal{F}_k\}_{k=0}^K$, satisfying*

$$\mathbb{E}[\boldsymbol{v}^k \mid \mathcal{F}_{k-1}] = \mathbf{0}, \qquad \|\boldsymbol{v}^k\| \leq a_k, \quad k = 1, \ldots, K. \tag{A.1}$$

Then for any $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\left\| \sum_{k=1}^K \boldsymbol{v}^k \right\| \leq 3\sqrt{\log(1/\delta) \sum_{k=1}^K a_k^2}. \tag{A.2}$$

The above bound is a standard vector-valued Azuma-Hoeffding concentration result.

References

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